



Original Article

Genome analysis reveals probiotic propensities of *Paenibacillus polymyxa* HK4

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ABSTRACT

The legislations on the usage of antibiotics as growth promoters and prophylactic agents have compelled to develop alternative tools to upsurge the animal protection and contain antibiotic usage. Probiotics have emerged as an effective antibiotic substitute in animal farming. The present study explores the probiotic perspective of *Paenibacillus polymyxa* HK4 interlinking the genotypic and phenotypic characteristics. The draft genome of HK4 revealed the presence of ORFs encoding the functions associated with tolerance to gastrointestinal stress and adhesion. The biosynthetic gene clusters encoding non-ribosomally synthesized peptides, polyketides and lanthipeptides such as fusaricidin, tridecaptin, polymyxin, paenilan and paenibacillin were annotated in HK4 genome. The strain harbored the chromosomal gene conferring the resistance to lincosamides. No functional gene encoding virulence or toxins could be identified in the genome of HK4. The genome analysis data was complemented by the *in vitro* experiments confirming its survival during gastrointestinal transit, antimicrobial potential and antibiotic sensitivity.

Nucleotide sequence accession number: The draft-genome sequence of *Paenibacillus polymyxa* HK4 has been deposited as whole-genome shotgun project at GenBank under the accession number PRJNA603023.

1. Introduction

Growing population and escalating wages are propelling an increasing demand for food globally. In India, the demand and consumption of milk, meat and egg are growing at a much faster rate as compared to cereals [1]. To fulfill this increasing demand for sustenance, the use of antibiotics as growth promoters in conventional livestock has been practiced. These antimicrobials are reported to enhance the animal growth through immunomodulation, thinning of the mucosal layer lining the gut and reduction of the bacterial load. Nevertheless, the exact mechanism behind growth promoting effect of antibiotics still remains ambiguous [2]. Despite of advantages associated with the use of antibiotics in animal husbandry, its excessive use is linked to the emergence of antibiotic resistant microbes, both in humans as well as in animals [3]. India is one of the top consumers of antibiotics in the animal agricultural sector, accounting for about 3% overall global consumption [4]. Currently, the information regarding antibiotic usage, resistance pattern and the laws imposed by the Indian government for animals is insufficient [4]. Although the use of growth promoting antibiotics are

forbidden to prevent emergence of drug resistant strains; it may lead to a decrease in productivity, higher disease incidence in birds and subsequent increase in usage of therapeutic antibiotics [2,5,6]. With this scenario in mind, alternatives to growth promoting antibiotics have become a prerequisite. Probiotics are enticing attention as one of the potential substitutes for antibiotics as growth promoters. They are defined as “live microorganisms that when administered in adequate amount confer a health benefit on the host” [7,8]. The available probiotics for animals largely include *Lactobacillus*, *Bifidobacterium*, *Streptococcus*, *Pediococcus*, *Enterococcus*, *Bacillus* genera and yeasts such as *Saccharomyces cerevisiae*. These are known to reduce the gastrointestinal stress, improve digestion, growth, feed efficacy, meat and egg quality, stimulate the immune system, reduce the serum cholesterol, maintain the normal intestinal microflora, inhibit the pathogens etc. [3,9–11]. The endospore forming bacilli are promising candidates in this arena owing to their stability and cost-effective benefits.

The members of genus *Paenibacillus*, ubiquitously distributed in nature, are gram positive endospore forming bacilli. These spore formers have an industrial prospect owing to their properties such as plant

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growth promotion, production of exopolysaccharides and extracellular enzymes, antimicrobial activity, nitrogen fixation and so on [12–16]. *Paenibacillus polymyxa* is one such non-pathogenic organism highly cited for its plant growth promoting activity. However, only few reports emphasize the probiotic potential of *P. polymyxa*. The *P. polymyxa* JB-0501, isolated from livestock feed supplement was explored as a probiotic candidate based on *in vitro* studies [17]. Similarly, the antibacterial and probiotic characteristics of *P. polymyxa* strain HGA4C, isolated from the gut of a fish, were demonstrated under *in vitro* and *in vivo* conditions [18]. *Paenibacillus* sp. Aloe-11 colonized the chicken intestine and assisted the degradation of non-starch polysaccharides with the help of secreted extracellular enzymes [19]. Also, one strain of *P. polymyxa* is being marketed as a probiotic for aquaculture under the trade name of Biostart® [20]. Although there are studies aiming to elucidate the *in vitro* and *in vivo* probiotic potential of *P. polymyxa*, no reports address the genome determinants associated with probiotic traits in this species.

The sequencing and functional annotation of genome has emerged as a rapid and cost-effective approach for evaluation of probiotic strains for its efficacy and safety [21]. Moreover, the information gained employing such analysis could help in the detection of molecular markers that may facilitate the screening and selection of probiotics [22]. The present study was therefore undertaken to correlate the phenotypic and genotypic probiotic traits as well as to investigate the safety of *Paenibacillus polymyxa* HK4.

2. Results

2.1. Genome attributes of *Paenibacillus polymyxa* HK4

The genome sequencing of *Paenibacillus polymyxa* HK4 using Illumina MiSeq sequencing platform generated a total of 2,11,6070 bp paired end reads with a sequencing depth of ~174 X. The *de novo* assembly of the obtained reads using MIRA assembler v4.0.2 generated 88 contigs constituting a circular chromosome of 6.08 Mb and 45.2% of average GC content. A total of 5601 protein coding genes (CDSs), 25 rRNA, 137 tRNA and 317 subsystems were predicted in the genome of HK4 (Table 1). The subsystem associated with carbohydrate metabolism, amino acid derivatives and protein metabolism were assigned high relative abundance in the genome of isolate HK4 (Supplementary Fig. S1). Out of 5601 predicted CDSs, 4175 CDSs were classified into 19 different cluster of orthologous groups (COGs) that were broadly categorized into a) information storage and processing; b) cellular processes and signaling; c) metabolism and d) poorly categorized genes. Amongst all, the significant portion (33.51%) was attributed to metabolism (Supplementary Table 1).

2.2. Mobile genetic elements

Mobile genetic elements (MGEs) are the DNA segments encoding enzymes and proteins that facilitate its mobility within the genome or

between the bacterial cells [23]. They mainly include insertion sequences, transposons, integrons, plasmids and integrative conjugative elements. Few of these elements were predicted in the genome of HK4. About 26 insertion sequences belonging to IS3, IS4, IS5, IS110, IS1182 and IS1585 families were identified in HK4 genome with the set threshold *E*-value of 0.0001 (Supplementary Table 2). Of the above mentioned IS elements, six insertion sequences from *Bacillus* sp. belonging to IS4 family exhibited maximum score bit (> 1000) with zero *E*-value. Using CRISPRCas finder, 9 CRISPR sequences and 3 associated genes (CAS) were identified in HK4 genome. A single intact prophage of 72.1 Kb size was identified in the genome of HK4. Genomic islands accounted for 8.96% of the HK4 genome. The majority of CDSs annotated as hypothetical proteins followed by transporters, phage proteins, polyketide synthase modules, beta-glucosidase (EC 3.2.1.21), beta-galactosidase (EC 3.2.1.23), insertion sequences, iron uptake system, etc. were mapped in 32 different genomic islands of HK4 contributing to the bacterium's fitness in its environmental niche [24,25] (Supplementary File 1).

2.3. Phylogenetic relationship

The use of 16S rRNA as a molecular marker for the determination of phylogeny limits the strain identification at the subgenus level and therefore, the multilocus sequence analysis (MLSA) was carried out to determine the phylogenies amongst the *Paenibacillus* sp. using an automated webserver. The phylogenetic tree constructed based on 80 house-keeping genes using neighbor joining method revealed that HK4 formed a separate clade with *Paenibacillus* sp. ICGB 2008 and *Paenibacillus polymyxa* A18, both isolated from the gut of insects, suggesting that it is more closely related to strains ICGB2008 and A18 (Fig. 1A). The phylogeny of HK4 was further established based on DNA-DNA hybridization and ANI indices. The DDH values of HK4 with *Paenibacillus* sp., ICGB 2008 and *Paenibacillus polymyxa* A18 were 91.60% and 91.50%, respectively, which ascertains that they are the members of the same species. Similarly, the genome relatedness of *Paenibacillus* sp. ICGB 2008 and *Paenibacillus polymyxa* A18 with HK4 was also assessed based on OrthoANI values that were found to be 98.97% and 98.99%, respectively, further confirming their relatedness (Table 2).

2.4. Molecular characterization of probiotic traits

The selection of any probiotic microorganism entails a systematic scheme that would eventually deliver a strain with maximum functional properties and minimum negative traits. The criteria for selection of any probiotic strain involves the determination of functional aspects such as its ability to endure the gastrointestinal transit i.e. the low pH of the stomach and the presence of bile in the intestine, pathogen exclusion or production of antimicrobial metabolites that may inhibit the growth of enteric pathogens, transient adherence to the intestinal epithelium, ability to improve feed digestibility and assessment of its safety [26–28].

2.5. Acid and Bile tolerance

The genome analysis of HK4 revealed the presence of a massive number of ORFs related to general and secondary stress, pH homeostasis and metabolic rearrangements, which may ensure its survival during the gastrointestinal transit. The ORFs encoding the proteins for F₀F₁-ATP synthase, Na⁺/H⁺ antiporter, amino acid decarboxylases (lysine, arginine, glutamate), alcohol dehydrogenase as well as lactate dehydrogenase, and agmatine deiminase (EC 3.5.3.12) were identified in the genome of HK4 (Table 3). Certain ORFs related to arginine deiminase system (viz. arginine deiminase, ornithine carbamoyltransferase (EC 2.1.3.3) (*arcB*), and arginine/ornithine antiporters (*arcD*)) and glutamate decarboxylase system (glutamate decarboxylase) were also mapped.

Owing to the lipophilic nature, the bile acids target the bacterial cell

Table 1

Genome attributes of *Paenibacillus polymyxa* HK4.

Attributes	<i>Paenibacillus polymyxa</i> HK4
Genome size	6,086,988 bp
GC content (%)	45.2
No. of contigs	88
No. of subsystems	317
N50	200,116
L50	11
Protein coding genes	5601
rRNAs	25
tRNAs	137
Plasmid	NA
Tandem repeats	92
CRISPR	9
Cas	3

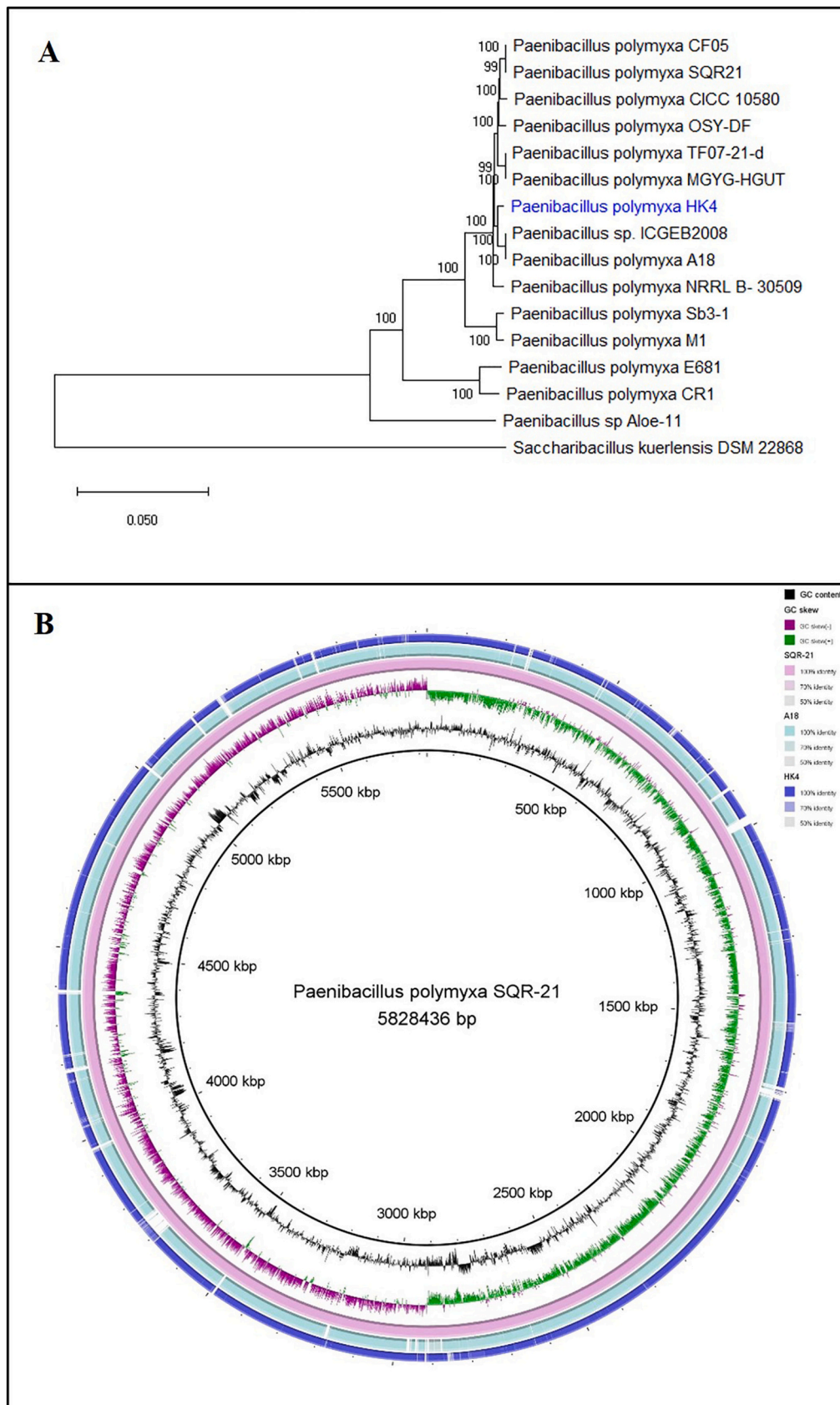


Fig. 1. Evolutionary analysis and comparison of *Paenibacillus polymyxa* HK4 with other *Paenibacillus* sp. (A) Dendrogram of HK4 based on Multilocus sequence analysis of 80 housekeeping genes using Neighbor joining method with 1000 bootstrap value constructed in MEGA X, (B) Schematic representation of HK4 genome. The rings from outside to inside represent the genome blastx hits of HK4 and A18 with SQR-21 as a reference genome followed by GC skew and GC content.

Table 2

In silico DNA-DNA hybridization (% DDH) and Average Nucleotide Identity (% ANI) values between *Paenibacillus polymyxa* HK4 and *Paenibacillus* sp. isolated from several sources.

Strain	Source	<i>In silico</i> DDH (%)	OrthoANI value (%)
<i>Paenibacillus polymyxa</i> SQR-21	Watermelon Rhizosphere	86.2	98.41
<i>Paenibacillus polymyxa</i> Sb3-1	Soil	60.8	94.94
<i>Paenibacillus polymyxa</i> A18	Termite gut	91.5	98.99
<i>Paenibacillus</i> sp. ICGEB2008	<i>Helicoverpa armigera</i> gut	91.6	98.97
<i>Paenibacillus polymyxa</i> OSY-DF	Fermented vegetable (Kimchee)	86.5	98.5
<i>Paenibacillus polymyxa</i> MGYG-HGUT-03687	Human gut	87.2	98.56
<i>Paenibacillus polymyxa</i> NRRL B-30509	Poultry environment	86.4	98.48
<i>Paenibacillus polymyxa</i> TF07-21-d	Faeces	87.2	98.56
<i>Paenibacillus polymyxa</i> CICC 10580	Noni fruits	86.5	98.48
<i>Paenibacillus polymyxa</i> EBL06	Leaf	86.6	98.47
<i>Paenibacillus polymyxa</i> M1	Wheat roots	59.9	94.83

membrane thereby affecting the viability of an organism [29]. Therefore, the microorganism must adapt several protective mechanisms to resist its deleterious action. The ORFs encoding bile and sodium symporter, glycine betaine ABC transport system and ornithine decarboxylase were detected in the genome of HK4 that might confer tolerance to bile. Besides this, the general stress response genes assist bacteria to respond and thrive in rapidly changing environmental conditions. The proteins involved in repair of macromolecules (DnaK/DnaJ chaperon, GroES, GroEL and Clp protease), sigma B regulation, secondary oxidative stress (catalase, thioredoxins, superoxide dismutase, peroxidase) were identified in the genome of HK4 (Table 3). In wet lab experiments, exposure to acidic environment (pH 2) lead to a decrease in average CFU/mL of HK4 from 2.18×10^7 CFU/mL to 2.5×10^4 CFU/mL (Fig. 2A). There was no significant decrease in its viability of HK4 at 0.2% bile whereas at 0.4% bile concentration in the growth medium, about 3 log reduction in viable cells of HK4 was observed (Fig. 2B).

2.6. Adhesion and aggregation

The property of adherence, often associated with aggregation, is considered a prerequisite for bacterial colonization in the gut environment. The transient colonization of a probiotic microorganism to the mucosa or the epithelial cells of the gastrointestinal tract implicates specific interactions between the bacterial cell surface and the host surface [30]. The genes encoding fibronectin/fibrinogen-binding protein, S-layer protein A domain (SlpA), MuBP domain, flagellin and lipoteichoic acid hook associated proteins that assist binding to extracellular matrix or mucosa were mapped in the HK4 genome. Also, sortase A, LPXTG specific, surface associated enolase, Clp protease, peptidyl-prolyl cis-trans isomerase (EC 5.2.1.8), protein associated with exopolysaccharides biosynthesis were identified in the genome of HK4. Moreover, several moonlighting proteins such as glyceraldehyde-3-phosphate dehydrogenase GAPDH (EC 1.2.1.12), elongation factor Tu, transaldolase (EC 2.2.1.2), and chaperone GroEL mediating bacterial adhesions were detected in HK4 (Table 3). The auto-aggregation percentage of HK4 at 2 h and 5 h of incubation was 16.4% and 31.5%, respectively. Similarly, the co-aggregation percentage of HK4 with *E. coli* at 2 h and 5 h of incubation was 12.5% and 19.9%, respectively and with *S. enterica* was 13.1% and 18.9%, respectively (Fig. 2C). The adherence percentage of HK4 in toluene, hexane and hexadecane was

Table 3

Acid, bile, stress, adhesion and aggregation related proteins encoded by the genome of *Paenibacillus polymyxa* HK4.

Property-Protein	Function
Acid Resistance	
Amino acid permease	Acquisition of exogenous amino acid
Glycine betaine ABC transport system	Osmoregulation
Metabolic rearrangements	
Alcohol dehydrogenase (EC 1.1.1.1)	Transfer electrons to electron transport chain (ETC)
Lactate dehydrogenase	Restores NAD^+ /NADH balance
Alanine dehydrogenase	Pumps proton out of cell (ETS)
Formate dehydrogenase	Pumps proton out of cell (ETS)
pH homeostasis	
F_0F_1 - ATP synthase	Proton translocation
Arginine decarboxylase (EC 4.1.1.19)	Proton consumption (converts arginine to agmatine)
Lysine decarboxylase	Proton consumption and production of basic amines
Agmatine deiminase (EC 3.5.3.12)	Converts agmatine to putresine releasing ammonia
Sodium hydrogen antiporter	Proton transporter
Arginine deiminase pathway (ADI)	
Arginine deiminase	Converts arginine to citrulline producing ammonia
Ornithine transcarbamylase	Converts citrulline to Ornithine and carbamyl phosphate
Arginine/Ornithine antiporter	Imports arginine and exports ornithine
Glutaminase and GAD system	
Glutaminase (EC 3.5.1.2)	Converts glutamine to glutamate
Glutamate decarboxylase and related	Converts glutamate to gamma-amino butyric acid
PLP-dependent proteins	Exchanges intracellular GABA with extracellular glutamate
Glutamate/GABA antiporter	
Bile resistance	
Bile acid sodium symporter	Bile acid uptake
Ornithine decarboxylase	Increases intracellular pH
ABC transporter permease protein	Transmembrane transport
General stress response	
Macromolecule/Protein repair chaperons	Refolding or degrading denatured protein, DNA repair and SOS response
σ^B regulation	Allows transcription of stress coding genes
Transcriptional regulators (heat stress)	–
Secondary oxidative response	Antioxidant defence molecules
Adhesion and aggregation	
Fibronectin/fibrinogen-binding protein	Facilitate binding to epithelial cells/extracellular matrix
Surface layer protein A domain	Regulation of immune response
Biofilm surface layer A	Facilitate binding epithelial cells extracellular matrix
Translation elongation factor Tu	
GroEL chaperon	Involved in host immunomodulation and pathogen aggregation
Pyruvate dehydrogenase E1 component beta subunit (EC 1.2.4.1)	Fibronectin binding protein
Sortase A, LPXTG specific	Bind to mucus
MucBP domain	Binds to mucus components
Glyceraldehyde-3-phosphate dehydrogenase	Adhesion to plasma components
Flagellar proteins (hook-associated protein FlgL, FlgK and cap protein FlgD)	Adhesion
Exopolysaccharide (EPS) biosynthesis protein	Synthesis of EPS
Enolase (EC 4.2.1.11)	Binds to plasma components
Peptidyl-propyl cis-trans isomerase (EC 5.2.1.8)	Pathogen aggregation

15.5, 22.7 and 4.2, respectively (Fig. 2D).

2.7. Antimicrobial activity

Once adhered to the gut, the probiotic produces and secretes the extracellular antimicrobial metabolites in the intestinal lumen that

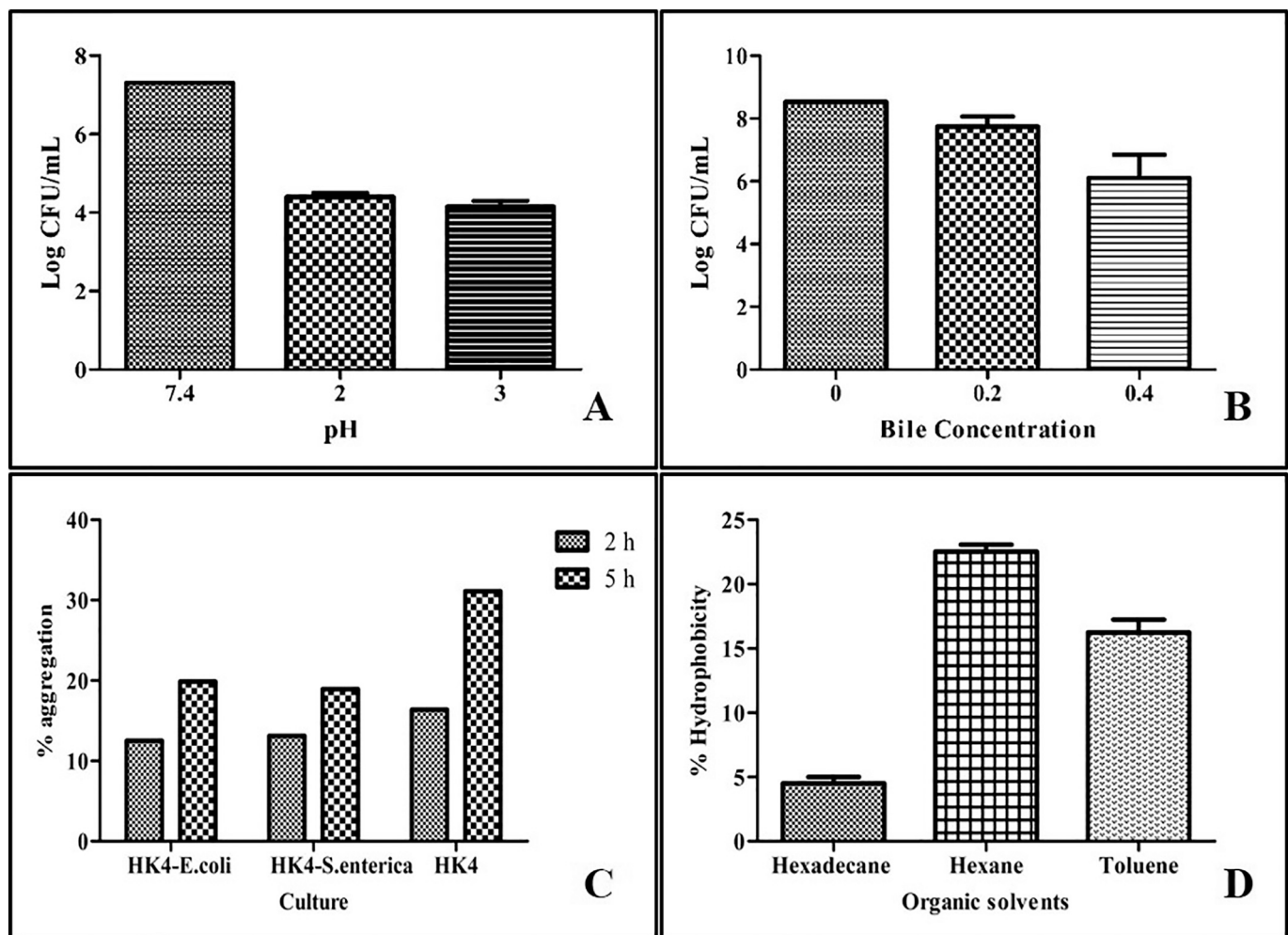


Fig. 2. Probiotic properties of HK4 under *in vitro* conditions (A) Survival of HK4 at different pH (B) Survival of HK4 at various bile concentration (C) Aggregation assay of HK4 (D) BATH assay of HK4.

might help it to compete with or suppress the pathogens. The bacterium HK4 exhibited antibacterial activity against all the tested gram positive and gram negative pathogens under *in vitro* conditions (Fig. 3). Moreover, HK4 when co-inoculated with pathogen in liquid media inhibited the growth of *E.coli* and *S. enteritidis* completely at 10 h and 24 h respectively. Several putative biosynthetic gene clusters accounting for 13.9% of the genome and responsible for the synthesis of secondary metabolites such as nonribosomal peptide synthetases (NRPS), polyketide synthases (PKS) and ribosomally synthesized and post translational modified peptides (RiPPs) were annotated in the draft genome of HK4 using antiSMASH web server. The NRPS gene clusters exhibiting 100% with similarity fusaricidin B, tridecaptin, anabaenopeptin and polymyxin gene clusters were traced in the genome of HK4 using antiSMASH. The RiPP biosynthetic gene clusters encoding for paenilan and paenibacillin, exhibiting similarity $\geq 90\%$ were also found in the genome (Fig. 4).

2.7.1. Safety assessment

One of the crucial selection criteria for any organism to be used as a feed additive is the concern about its safety, particularly when fed to humans or animals. The important safety related aspects for a probiotic include the assessment of its antibiotic resistance and pathogenicity determinants (virulence and toxin production).

2.7.1.1. Antibiotic resistance determinants. The animal gut microbiota is considered as one of a crucial reservoir of antibiotic resistant (AR) genes.

Probiotics harboring resistance genes are themselves not considered to be pernicious but the dissemination of resistance genes from probiotic to existing gut microflora and the opportunistic pathogens sharing the common intestinal niche may pose serious ramifications [31]. Therefore, the screening of antibiotic resistance determinants for direct-fed bacterial products is essential. The genome analysis of HK4 for the presence of antibiotic resistance genes using CARD revealed the presence of Llm 23S ribosomal rRNA methyltransferase gene family conferring resistance to clindamycin by altering the antibiotic target. The antibiotic susceptibility testing of HK4 to 19 different antibiotics was experimentally verified using disc diffusion assay. HK4 was found to be sensitive to all the analyzed antibiotics except bacitracin, oxacillin and polymyxin B (Supplementary Table 3). However, it was found to be less sensitive towards clindamycin and cephalothin (Supplementary Fig. S2).

2.7.1.2. Virulence determinants. Several putative virulence and toxin encoding genes associated with safety were evaluated in the genome of HK4 using Virulence Factor Database (VFDB) with threshold coverage of $>70\%$, percent similarity $>70\%$ and *E*-value <0.0001 as well as employing Virulence finder v2.0 with 85% threshold and 60% sequence similarity [32]. No virulence genes were identified in the genome of HK4 using Virulence finder. A total of 19 genes putatively related to virulence were predicted by VFDB and are summarized in Supplementary Table 4. According to COG database, these genes are associated with the basic cellular functions or defence such as carbohydrate transport and

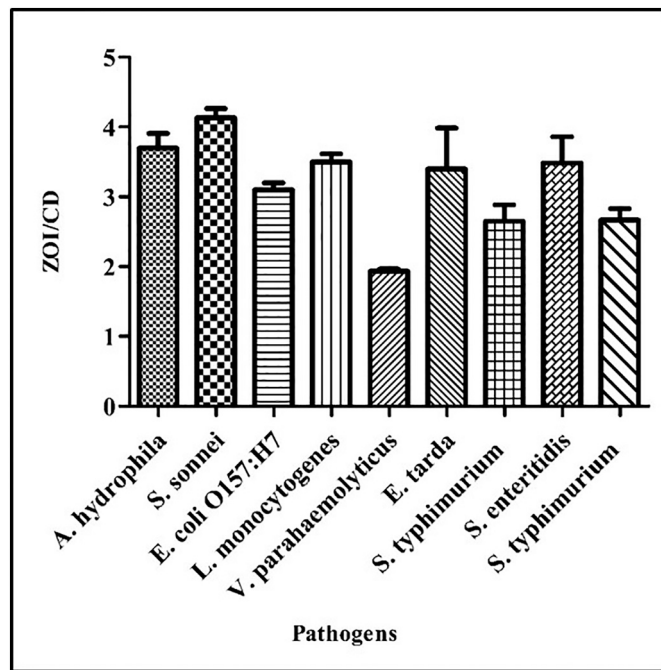


Fig. 3. Antibacterial activity of *Paenibacillus polymyxa* HK4 against different pathogens (ZOI/CD refers to ratio of zone of inhibition to colony diameter).

metabolism (G), cell wall/membrane/envelope biogenesis (M), cell motility (N), intracellular trafficking, secretion, and vesicular transport (U), posttranslational modification, protein turnover, chaperones (O), secondary metabolites biosynthesis, transport and catabolism (Q) and defence mechanisms (V). The genes predicted using VFDB may be termed as virulence factors in the case of pathogens to survive the physiological stresses, however the same factors could favor a symbiotic/commensal association of a microbe with the host [33]. It must be noted that all hits obtained for virulence factors in our study exhibited <70% coverage, which is below the threshold. No hemolytic activity was obtained on 5% sheep blood agar plates supporting the safety of HK4.

2.8. Carbohydrate-active enzymes identification

HK4 was screened for the production of its extracellular enzymes using substrate hydrolysis method. It displayed the hydrolysis of cellulose, chitin, xylan, pectin, starch and casein with maximum ZOH/CD

(zone of hydrolysis by colony diameter ratio) ratio for chitin, followed by xylan, cellulose and pectin (Supplementary Fig. S3). The carbohydrate utilization profile of HK4 was tested using 35 different carbohydrate sources. HK4 was found to utilize 19 carbohydrates viz. lactose, xylose, maltose, fructose, dextrose, galactose, raffinose, trehalose, melibiose, sucrose, L-arabinose, mannose, glycerol, salicin, mannitol, cellobiose, ONPG, esculin and D-arabinose (Supplementary Fig. S4). The genes involved in the utilization of carbohydrate substrates are summarized in Supplementary Table 5.

The screening of protein coding ORFs obtained using Pfam, RAST, and HMMscan in the genome of HK4 for the presence of carbohydrate active enzymes using dbCAN2 meta server revealed a large number of enzymes involved its metabolism. The study revealed the dominance of glycoside hydrolases family of enzymes (52), followed by Glycosyl transferase (12), carbohydrate esterases (10), pectate lyase (7) and enzymes with auxiliary activities (3) in HK4. Besides, the carbohydrate-binding modules (20) that enhance the activity of carbohydrate active enzymes were also detected in HK4.

The identified CAZymes families were then mined for the detection of enzymes associated with the breakdown of carbohydrates and complex substrates viz. chitin, xylan, pectin, cellulose and starch. In HK4, the glycoside hydrolase family (GH) 13 encoding α -amylase (EC 3.2.1.1) and oligo- α -glucosidase (EC 3.2.1.10) involved in the breakdown of starch, GH43 involved in xylan breakdown (β -xylosidase EC 3.2.1.37, xylanase EC 3.2.1.8) and GH5 (*endo*- β -1,4-glucanase/ cellulase EC 3.2.1.4; *endo*- β -1,4-xylanase EC 3.2.1.8, β -glucosidase EC 3.2.1.21) catalyzing the hydrolysis of both cellulose and xylan were detected. Besides GH, carbohydrate esterase (CE) also catalyzes the degradation of polysaccharides. CE4 involved in deacetylation of xylan (acetyl xylan esterase EC 3.1.1.72) and chitin (chitin deacetylase EC 3.5.1.41) were found to be higher in HK4 followed by CE10, active against non-carbohydrate substrates. Polysaccharide lyases help in the cleavage of activated glycoside linkages in acidic polysaccharides. HK4 mainly consisted of PL1 that plays a role in pectin degradation. Auxiliary amylases (AA) are the redox active catalytic modules that act in conjunction with the CAZymes. In HK4, three different families of AA namely AA1, AA6 and AA7 involved in the biodegradation of aromatic compounds and biotransformation of lignin-cellulosic compounds were determined. The CAZymes involved in the degradation of complex substrates are summarized in Table 4. Glycosyl transferases are known to play a role in the synthesis of complex carbohydrates and glycoconjugates essential for the bacterial adhesion, signaling process and cell wall biosynthesis. In HK4, the enzymes belonging to glycosyl transferase families (GT2 and GT4) were found to be abundant. The GT2 family of enzymes is involved in the formation of β -linked polysaccharides such as cellulose, chitin and hyaluronan and GT4 consists of enzymes acting on nucleotide sugar and

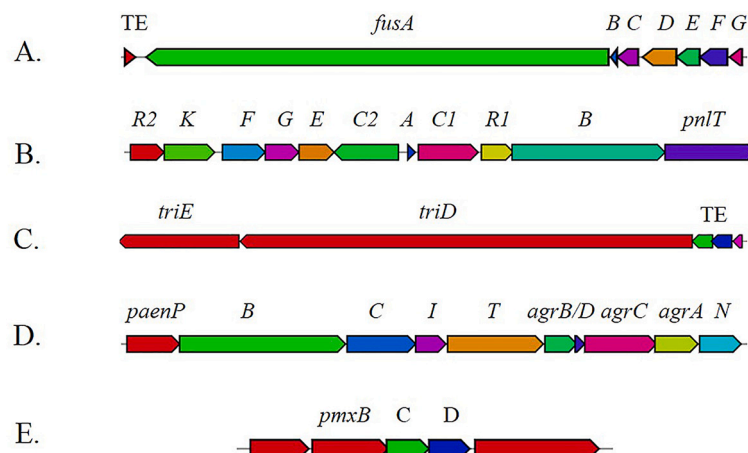


Fig. 4. Putative gene cluster for biosynthesis of secondary metabolites (A) Fusaricidin, (B) Paenilan, (C) Tridecaptin, (D) Paenibacillin, (E) Polymyxin.

Table 4

CAZymes involved in hydrolysis of complex substrate (cellulose, starch, xylan, pectin, chitin and cellulose).

Substrate	Protein	CAZymes Family
Starch	Alpha-amylase (EC 3.2.1.1)	GH13, GH14, GH31
	Beta-amylase (EC 3.2.1.2)	CBM26, CBM34
	Maltodextrin ABC transporter, ATP-binding protein MsmX	
	Maltose/maltodextrin ABC transporter, permease protein MalF	
Xylan	Oligo-1,6-glucosidase (EC 3.2.1.10)	
	Maltodextrin glucosidase (EC 3.2.1.20)	
	Endo-1,4-beta-xylanase (EC 3.2.1.8)	GH3, GH5, GH10, GH11,
	Xylan 1,4-beta-xylosidase (EC 3.2.1.37)	GH43, GH51, GH52, GH67,
	Xylose-responsive transcription regulator, ROK family	GH115, GH141
	Xyloside transporter XynT	CE1, CE2, CE3, CE4, CE7,
	Xylose ABC transporter, substrate-binding component	CE12
	Xylose ABC transporter, permease component 2	CBM22, CBM36, CBM59,
	Ribose ABC transporter, substrate-binding protein RbsB (TC 3.A.1.2.1)	CMB35, CBM54
	Ribose ABC transporter, ATP-binding protein RbsA (TC 3.A.1.2.1)	
	ABC transporter, substrate-binding protein (cluster 2, ribose/xylose/arabinose/galactose)	
	Ribose ABC transporter, permease protein RbsC (TC 3.A.1.2.1)	
	Xylulose kinase (EC 2.7.1.17)	
	Xylose isomerase (EC 5.3.1.5)	
	Xylan alpha-1,2-glucuronosidase (EC 3.2.1.131)	
	Feruloyl esterase (EC 3.1.1.73)	
	Acetyl xylan esterase	
	Xylanase, glycosyl hydrolase family 10	
Pectin	Putative pectin degradation protein	PL1, PL3, PL9, PL10
	Pectin lyase (EC 4.2.2.10)	
	Polygalacturonase (EC 3.2.1.15)	
	Uronate isomerase (EC 5.3.1.12)	
	Altronate oxidoreductase (EC 1.1.1.58)	
Chitin	Altronate dehydratase (EC 4.2.1.7)	
	Chitinase (EC 3.2.1.14)	GH18, GH23, GH48, GH85
	PTS system, N-acetylglucosamine-specific IIC component / PTS system, N-acetylglucosamine-specific IIB component (EC 2.7.1.193) / PTS system, N-acetylglucosamine-specific IIA component (EC 2.7.1.193)	CBM50, CBM5
	N-acetylglucosamine-6-phosphate deacetylase (EC 3.5.1.25)	
	Glucosamine-6-phosphate deaminase (EC 3.5.99.6)	
	Endoglucanase celA precursor (EC 3.2.1.4) (Endo-1,4-beta-glucanase) (Cellulase)	
Cellulose	beta-glucosidase (EC 3.2.1.21)	GH1, GH3, GH5, GH6,
	Cellobiose phosphorylase (EC 2.4.1.20)	GH10, GH12, GH44, GH48,
	PTS system, cellobiose-specific IIB component (EC 2.7.1.205)	GH51, GH85
	6-phospho-beta-glucosidase (EC 3.2.1.86)	CBM59, CBM46, CBM44,
	Cellobiohydrolase A (1,4-beta-cellobiosidase A)	CBM63, CBM16,
	Cellulose 1,4-beta-cellobiosidase (reducing end) (EC 3.2.1.176)	

lipid-phospho-sugar donors.

3. Discussion

The portfolio of antibiotics used for therapeutic, sub-therapeutic and prophylaxis in the animal sector is alarming with the consequences for animal and human health. For the sustainable development of animal

agriculture, the alternatives to antibiotics are urgently required and amongst them probiotics hog the limelight. The bacterium, *P. polymyxa*, ubiquitously distributed in nature, is well known for its plant growth promoting activity. They have the potential to produce wide range of antimicrobial compounds that might be active against plant, animal and human pathogens [13,34]. However, there are no reports emphasizing the probiotic potential of *P. polymyxa* in conjunction with genome studies. The present investigation reports the screening of probiotic potential of *P. polymyxa* HK4 by employing both wet lab investigation of phenotypic traits as well as genomics based *in silico* prediction of probiotic traits. *Paenibacillus polymyxa* HK4, isolated from the roots of *Epipremnum aureum* plant consists of 6.08 Mbp circular chromosome. The genome comparison based on phylogenetic analysis, *in silico* DDH and ANI depicted that HK4 is more closely related to *P. polymyxa* A18 and *Paenibacillus* sp. ICGB2008, isolated from insect gut, in comparison to other *Paenibacillus* sp. reported from plant rhizosphere (Table 2). It is noteworthy to mention here that at genome level, HK4 exhibited more than 86% DDH and 98% ANI with several *P. polymyxa* isolated from the human gut or poultry environment (Table 2).

The capability of an organism to reciprocate to the environmental variations ensures its survival in the niche. The probiotic bacteria need to survive the transit and complexity of the gastrointestinal tract. Therefore, the tolerance to acid/bile and the associated extrusion mechanisms are the desirable traits to be investigated. In HK4, the probable mechanism that may confer resistance to acid stress involves the activation of sigma B gene, a molecular biomarker to track moderate acid resistance, controlled by expression of conserved RsbVW partner switching [35] (Table 3). Under normal physiological conditions, sigma B is in inactive form due to its association with anti-sigma factor RbsW. On exposure to stress conditions, RbsV, an anti-sigma factor antagonist binds to RbsW forming RbsV-RbsW complex leading to the release of sigma B and therefore the transcription of genes associated with it. Besides sigma B gene, the gene encoding catalase linked to oxidative stress is reported as a biomarker to track high acid resistance and therefore may assist acid tolerance in HK4 [35]. The bacterial exposure to stress conditions may lead to metabolic rearrangements and induction of alternative ETC components resulting in secondary oxidative stress [36]. The proteins viz. GroEL, Clp, RecA and DnaK, indirectly protect the bacteria against acid stress by ensuring the proper functioning of the DNA repair and recombination pathways, were found in HK4 genome (Table 3). Secondly, the enzymes that may contribute to the maintenance of pH homeostasis in HK4 include F₀ F₁ - ATP synthase and Na⁺/H⁺ antiporter, acting as channels for proton translocation at the cost of ATP synthesis or ATP hydrolysis. In gram positive bacteria, the proton consuming reactions such as glutamate decarboxylation plays a role in the maintenance of pH homeostasis [37]. Although the gene encoding glutamate decarboxylase enzyme was found in the genome of HK4, however the glutamate/GABA symporter essential for the activity of GAD system was absent in HK4 implying that it may not assist in acid tolerance. The enzymes decarboxylases and dehydrogenases found in HK4 are known to contribute to pH homeostasis by consuming the protons as a part of decarboxylation and fermentation reactions, respectively. Also, the amino acid decarboxylases found in HK4 are reported to generate basic amines as one of the products resulting in the increase of extracellular pH [35]. A similar set of genes conferring resistance to acid were annotated in the genome of *Bacillus coagulans* HS243 [38].

The bile acids act by disrupting the bacterial membrane, denature the proteins, cause oxidative damage to the DNA, chelate iron and calcium and govern the expression of the eukaryotic genes involved in host defence and immunity [39]. To resist the deleterious action of these compounds the microorganism must have evolved specific defence mechanism such as bile efflux, induction of general stress proteins, bile salt hydrolysis and reorganization of central metabolic pathways [40–42]. The genes coding bile and sodium symporter, glycine betaine ABC transport system and ornithine decarboxylase detected in the

genome of HK4 might help the bacterium to thrive in presence of bile [43,44]. Bile salts in presence of bile salt hydrolase enzymes split into unconjugated bile acids and glycine/taurine. These unconjugated bile acids being weak acids can passively diffuse into the cell and hence can cause intracellular acidification. Arginine decarboxylase (EC 4.1.1.19) /Ornithine decarboxylase enzyme catalyzes the decarboxylation of arginine/ornithine to putresine, a biogenic amine, that leads to an increase in pH [45,46]. The F_0F_1 -ATP synthase translocates the protons out and is therefore considered as a connecting link between acid and bile tolerance as it sustains intracellular pH under the bile environment as well [35,40,47]. The genome studies of *Bacillus clausii*, a commercially available probiotic and other bacilli species have suggested the role of F_0F_1 -ATP synthase, ornithine decarboxylase, bile salt sodium symporter and stress response genes to counteract the toxic effects of acid and bile [43,44].

Once inside the gut, the next step is transient colonization of probiotic to the epithelial cells of the gastrointestinal (GI) tract. The mechanism of adherence involves the binding of surface associated molecules to the mucus layer lining of the gut or the epithelial cells. The mucin binding protein (MucBP) domains and lipoteichoic acid are the bacterial surface adhesion proteins that bind to the mucin and glycans of the epithelial cells mediating the host-microbe interactions in the GI tract [22,48,49]. These proteins are characterized by LPXTG motif on their surface and are known to be cleaved by sortase enzyme, another protein identified in HK4, before being projected on peptidoglycan layer [22,49–51]. Such proteins may assist its adhesion to the GI tract. The fibronectin binding protein Fbp and enolase found in HK4 are known to bind fibronectin of the extracellular matrix and collagen, respectively [52,53]. The putative role of fibronectin binding protein in adhesion to the extracellular matrix of epithelial cells has been described through genome studies in *Lactobacillus gasseri* ATCC 33323 [54]. Provencio et al. documented the importance of FbpA in adhesion, by constructing *fbpA* mutant which exhibited a lower binding capacity to immobilized fibronectin as well as the reduction in bacterial surface hydrophobicity as compared to wild type [53]. Several proteins associated with flagella such as FlgL, FlgK and FlgD were found in HK4. Flagellum, the locomotory organelle with thin filaments extending from the surface of bacteria can function as adhesin owing to flagellin protein and facilitate adherence to the hydrophobic surface of intestine such as mucin [55–57]. Sanchez and his coworkers demonstrated the role of flagellin in adherence to mucin and fibronectin by using a recombinant *Lactococcus* strain expressing the flagellin of probiotic *B. cereus* CH cloned in lactococcal vector pNZ8110 [58]. The ORFs associated with exopolysaccharide biosynthesis and biofilm genesis regulatory protein (LuxS), mapped in the genome of HK4, may promote the prolonged colonization in the intestine preventing the adherence of pathogens. Similar proteins, involved in adhesion were found to be upregulated in *Bifidobacterium* conferring resistance to bile salts and thereby depicting the association between bile tolerance and adhesion [40,59]. The moonlighting proteins with an essential role in microbial growth and metabolism are known to reversibly associate with the intestinal mucosa at acidic pH without displaying any extracellular motif or domains [50,60,61]. The acidic pH of the intestinal mucosa confers a positive charge to such proteins enabling them to bind to the negatively charged lipoteichoic acid found in bacterial membrane with concomitant release at neutral or alkaline pH of the intestine. Although many proteins involved in adhesion and biofilm formation are associated with the virulence of few pathogens, their presence in probiotic cultures may serve as a means of positive selection increasing their adhesion ability to host surface favoring pathogen exclusion [49,62,63].

A probiotic with good adhesion capability can compete with the pathogens for host binding sites. Competitive exclusion, co-aggregation, production of antimicrobial agents and immune system stimulation are the mechanisms of probiotic for pathogen's elimination [21]. Several biosynthetic gene clusters related to the production of antimicrobial metabolites were identified in the genome of HK4 comprising 13.9% of

the genome. The NRPS gene clusters in HK4 included fusaricidin B, polymyxin B, tridecaptin and anabaenopeptin. A ~31.5 Kb gene cluster of fusaricidin was detected with *fusF* encoding for acyl-CoA ligase, *fusC* for 3-oxoacyl-acyl carrier protein synthase, *fusA* for fusaricidin synthetase, *fusB* encoding (3R)-hydroxymyristoyl-acyl carrier protein dehydratase, *fusD* encoding acetolactate synthase large subunit, *fusG* encoding enoyl-acyl carrier protein reductase and *fusTE* for alpha/beta hydrolase-thioesterase. Fusaricidin is a cyclic depsipeptide antibiotic specifically produced by *Paenibacillus polymyxa* with potent antifungal activity and antibacterial activity against gram positive bacteria [15,64,65]. Another biosynthetic gene cluster in HK4, tridecaptin extends over ~53.5 Kb and contains ORFs with two encoding for non-ribosomal peptide synthetase (*triD*, *triE*) and *triA* encoding thioesterase. The genome mining of *Paenibacillus polymyxa* E681 characterized tridecaptin as a linear tridecapeptides active against gram negative bacteria and multidrug resistant pathogens such as *Klebsiella pneumoniae* and *Acinetobacter baumannii* [66]. Polymyxin representing lipodecapeptides are the principal compounds of *Paenibacillus polymyxa* active against gram negative bacteria and a few gram positive bacteria [67]. The incomplete gene cluster of polymyxin in HK4 (~14.7 Kb) consisting three complete ORFs (*pmxB*, *pmxC* and *pmxD*) and two incomplete ORFs viz., *pmxA* (2.4 Kb) and *pmxE* (5.4 Kb) was annotated (100% identity) to contig no. 43 (14.7 Kb) of the HK4 draft genome. This missing sequence of polymyxin gene cluster in HK4 draft genome corresponds to the gaps in the draft sequence owing to the shortcoming of sequencing output. Nevertheless, it reflects the polymyxin producing potential of HK4, which needs to be confirmed experimentally. According to Choi et al., polymyxin gene cluster (40.6 Kb) consists of five ORFs, three encoding polymyxin synthetase gene (*pmxA*, *pmxB* and *pmxE*) and two encoding membrane transporters (*pmxC* and *pmxD*) exhibiting homology with ABC transporters of *Brevibacillus brevis* [68]. It is worthy to mention here that polymyxin B has been demonstrated to be effective in treatment of infections caused by major multidrug resistant bacteria such as *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, and *Klebsiella pneumoniae* [69]. The ribosomally synthesized and post-translationally modified peptide (RIPP) encoding gene clusters in HK4 include paenilan and paenibacillin, both belonging to class lantibiotics and active against gram positive bacteria. The paenibacillin gene cluster consists of ORFs encoding proteins for transport (*paenT*), protease (*paenP*, agr quorum sensing (*agrA*, *agrB*, *agrC*, *agrD*), modification (*paenB*, *paenC*, *paenN*) and immunity (*paenI*). Similar gene cluster is reported in *Paenibacillus polymyxa* OSY-DF [70]. Similarly, the biosynthetic gene cluster of paenilan consists of eleven open reading frame encoding paenilan transporter (*pnlT*), immunity (*pnlE*, *pnlG*, *pnlF*), modification (*pnlB*, *pnlC1*, *pnlC2*), regulation (*pnlR1*, *pnlR2*, *pnlK*) and synthesis (*pnlA*) [71]. The presence of varied biosynthetic gene clusters in HK4 might be responsible for its broad spectrum antimicrobial activity observed in laboratory assays (Fig. 3).

One of the essential requirement of any bacterial culture to be used as a probiotic or feed additive is its safety. The antibiotic resistance in bacteria can be intrinsic or may be acquired through horizontal gene transfer (HGT) or mutation. The animal gut is considered as an open environment for the transfer of antibiotic resistance amongst the intestinal microflora through HGT however, the intrinsic resistance has a minimum potential of dissemination [31]. These resistant bacteria can be transmitted to humans through animal waste or products polluting an entire food chain. According to the European Food Safety Assessment (EFSA), any microorganism to be selected as a feed additive or direct-fed microbial should be least resistant to the antibiotics of human and veterinary importance [72]. The investigation of HK4 genome revealed resistance to the lincosamide class of drugs associated with the modification of target site. These predictions coincide with the experimental results. HK4 was sensitive to all the tested antibiotics except bacitracin, oxacillin, cephalothin, polymyxin B and clindamycin. This depicts the safety of the strain and reduces the risk of antibiotic resistance gene transfer to native gut flora of the host.

Another safety aspect *i.e.* the virulence factors, refers to the gene products produced by a microorganism enabling it to colonize the host niche, evade the host immune response and lead to tissue damage [73]. Virulence factors include several surface associated structures such as capsular polysaccharides, outer membrane protein and lipopolysaccharides, secretion system, siderophores, catalases, hemolysin or secreted toxins that help bacteria to establish infection [21,74]. Classic virulence factors such as adhesins or stress proteins associated with the commensal bacteria are referred as host-interaction factors. Adhesion proteins such as mucin binding, collagen binding proteins, gram positive anchor protein (LPXTG) and fibronectin binding protein (FBP) were reported in the genome based investigation of two market probiotics *viz.*, *Bacillus coagulans* S-lac and *B. subtilis* TO-A-JPC [75]. Similarly, flagellar hook proteins, type IV prokaryotic filamentous adhesins and FBP were reported in the soil isolate *Bacillus coagulans* LBSC (DSM 17654) assessed as a safe probiotic [76]. The stress related proteins such as Clp protease, DnaK, GroEL, GroES along with capsular polysaccharide biosynthesis proteins Cps4, heat shock protein Hsp and sortase protein, annotated in HK4 genome are also present in the probiotic *Lactobacillus helveticus* MTCC 5463 [32]. Few commensals inhabiting the animal gut secrete siderophore for iron acquisition and contribute to the pathogen exclusion process and confine pro-oxidant mucosal responses thereby alleviating proinflammatory responses [77]. Several siderophore genes such as enterobactin synthase, pyoverdine synthetase and siderophore efflux system were detected in the HK4 genome. The siderophores, bacillibactin and pyoverdine were shown to be produced by *Bifidobacterium* species using genomics as well as proteomics and were corroborated to its adaptation under iron limiting environment of the GI tract [78]. The enterobactin, a catechol type siderophore is reported as an effective anti-cancer compound depriving the availability of iron to cancer cells [79].

The animal feed generally comprises a mixture of plant and animal based ingredients. These feed ingredients *viz.* starch, protein, lipids may be digested by the enzymes produced in the digestive tract of animals, however components such as non-starch polysaccharides (NSPs) *e.g.* cellulose, chitin, pectin, glucans, *etc.* may remain undigested [80]. The intestinal microflora and the administered probiotics play an pivotal role in effective digestion and absorption of feed ingested by the host owing to the number of extracellular enzymes (*e.g.* cellulase, chitinase, xylanase, pectinase, amylase and protease) produced by them. Glycoside hydrolases, a major class in HK4 genome, are the enzymes catalyzing the hydrolysis of relatively complex polysaccharides such as starch, cellulose, chitin and xylan into simpler products. The microorganisms co-producing xylanase and cellulases owe inclusive commercial applications [81]. The use of xylanase, cellulase and pectinase decreases the feed viscosity and enhances nutrient absorption. The cellulose can be degraded into its monomeric form by the synergistic activity of endo-1, 4- β -glucanase (EC 3.2.1.4), GH1 β -glucosidase (EC 3.2.1.21), cellobiose phosphorylase (EC 2.4.1.20) and 6-phospho- β -glucosidase (EC 3.2.1.86) all of which are present in HK4. The endocellulases and exocellulases belong to the GH families GH1, GH3, GH5, GH6, GH10, GH12, GH44, GH48, GH51 and GH85. Besides GH, the CBMs that promote the enzyme-substrate interaction assisting cellulose hydrolysis are also detected in HK4 genome such as CBM16, CBM44, CBM46, CBM59 and CBM63. Similarly, for hydrolysis of xylan, GH detected in HK4 genome includes GH3, GH5, GH10, GH11, GH43, GH51, GH52, GH67, GH115 and GH141. Besides GH, acetyl xylan esterase (CE) catalyzing the degradation of xylan in HK4 belongs to CE1, CE2, CE3, CE4, CE7, CE12 and CBMs with CBM22, CBM36, CBM59, CBM35, CBM54 families. The major GH enzymes that may catalyze xylan hydrolysis in HK4 include xylan 1, 4- β -xylosidase (EC 3.2.1.37), xylose isomerase (EC 5.3.1.5), xylose kinase (EC 2.7.1.17), endo-1, 4- β -xylanase (EC 3.2.1.8) and xylan 1, 4- β -xylosidase (EC 3.2.1.37). The study carried out by Mathlouthi et al. demonstrated the synergistic action of xylanase and β -glucanase in effective hydrolysis of soluble non-starch polysaccharides [82]. Chitin, a polymer of β -1, 4-*N*-acetylglucosamine is a component of

fungi, insects, shrimps and lobsters. Enzymes that might be involved in chitin degradation in HK4 involve chitinase (EC 3.2.1.14), *N*-acetylglucosamine-6-phosphate deacetylase (EC 3.5.1.25) and glucosamine-6-phosphate deaminase (EC 3.5.99.6). GH and CBM families identified in HK4 genome for chitin degradation include GH18, GH23, GH48, GH85 and CBM5, CBM50, respectively. Similarly, pectin hydrolysis enzymes predicted in HK4 genome are pectin lyase (EC 4.2.2.10), polygalacturonase (EC 3.2.1.15), uronate isomerase (EC 5.3.1.12), altronate oxidoreductase (EC 1.1.1.58), altronate dehydratase (EC 4.2.1.7) and pectinesterase (EC 3.1.1.11) belonging to polysaccharide lyase families PL1, PL3, PL9 and PL10. The analysis of a soil isolate *Bacillus subtilis*, selected based on enzymatic potential, demonstrated better feed efficiency and growth performance in broiler chicken [3] thereby elucidating a criterion for probiotic selection. Furthermore, the predicted transport proteins and enzymes involved in uptake and utilization of different carbohydrates imparts biochemical fitness to HK4 for survival in gut ecosystem and therefore may assist digestion of complex carbohydrates present in animal feed.

4. Conclusion

The present study was undertaken to demonstrate the probiotic intricacies of *Paenibacillus polymyxa* HK4. In particular, the traits essential for a probiotic to thrive through the gastrointestinal transit and transiently colonize the gut were determined at molecular as well as physiological level. Moreover, the passel of antimicrobials and enzymes harnessed by the strain can mitigate the intestinal pathogens and assist the breakdown of complex macromolecules enmeshed in animal diet. This study unveiled an inherent probiotic prospective of HK4 offering a viable counterfeit of antibiotics in animal agriculture and industries. Moreover, the present study can act as a framework to further explore the genome determinants of *Paenibacillus polymyxa* as a probiotic.

5. Materials and methods

5.1. Growth conditions and DNA extraction

The bacterial culture, *Paenibacillus polymyxa* HK4, was isolated from the roots of *Epipremnum aureum* plant collected from Anand, Gujarat, India. The strain HK4 was maintained at -20°C in 20% (v/v) glycerol and for experiments the inoculum was prepared by sub-culturing it on sterile tryptone soya agar media (TSA, HiMedia, India) followed by incubation at 37°C . For the isolation of genomic DNA, the strain was cultured in a sterile TSB (tryptone soya broth) and incubated at 37°C (150 rpm). The genomic DNA of HK4 was extracted from mid exponential phase culture using CTAB method [83].

5.2. Genome sequencing, assembly and annotation

The quality and the quantity of an isolated genomic DNA were evaluated using agarose gel electrophoresis (0.8%) and Qubit (ThermoFisher, USA) respectively. The genome sequencing libraries were prepared using Nextera™ DNA Flex library preparation kit and the constructed libraries were then sequenced on an Illumina MiSeq platform (Illumina, Inc. San Diego, CA) with paired end technology (2×250 bp) using Illumina V2 reagent kit. The quality filtered reads ($Q \geq 30$) were assembled using MIRA v4.0.2 assembler [84]. The gene annotation of the assembled genome was carried out using Rapid Annotation Subsystems Technology (RAST) [85]. The ORFs were identified using Prodigal [86] followed by HMMScan [87] and DIAMOND blast [88] with NCBI nr [89] and Pfam (<https://pfam.xfam.org>) as reference databases. In addition, the functional gene annotation also carried out using COG [90] and KEGG database [91]. The tRNA and rRNA prediction was carried out using tRNA scan-SE 1.21 and RNAmmer respectively [92,93]. The complement of proteins encoded by the genome were classified into cluster of orthologous group using eggNOG

with the help of OmicsBox software v1.2.4 (www.biobam.com/omicsbox). CRISPRCas finder tool was used to identify the CRISPR/Cas genes. The presence of tandem repeats and insertion elements were identified using ISfinder [94] and tandem repeats finder [95]. The prophage sequences within the genome were investigated using (PHASTER) PHAge Search Tool – Enhanced Release [96]. The genomic islands in the genome were predicted using IslandViewer4 [97]. The secondary metabolites gene clusters were identified using antiSMASH v5.0 [98]. The presence of putative virulence genes was predicted using Virulence Factor of Bacterial Pathogen database (VFDB) [99] and VirulenceFinder v2.0 [100]. The pattern of antibiotic resistance was investigated using Comprehensive Antibiotic Resistance Database (CARD) [101]. A Blast atlas was constructed using BLAST Ring Image Generator (BRIG) [102] by performing blastx of strains HK4 and A18 against *Paenibacillus polymyxa* SQR-21, an isolate from watermelon rhizosphere. The genome sequence is deposited in GenBank using NCBI Prokaryotic Genome Annotation Pipeline (PGAP).

5.3. Phylogenetic analysis

The whole genome phylogeny of *Paenibacillus polymyxa* HK4 was analyzed using an automated webserver (<https://automl.st.ziemertlab.com>) with *De novo* workflow mode. The concatenated aligned sequences retrieved from the webserver were imported in MEGA X software [103] for phylogenetic tree construction using neighbor joining method. The phylogenetic tree was constructed using the predicted Tamura Nei model in association with gamma distribution. Also, *in silico* genome to genome distance of HK4 with selected closely related *Paenibacillus* sp. was investigated using Genome to genome distance calculator web server v2.1 [104] that takes into account Genome BLAST Distance Phylogeny (GBDP) and *in silico* DNA-DNA hybridization. The overall genome relatedness of HK4 with reported *Paenibacillus* sp. isolated from several sources was carried out using an average nucleotide identity calculator [105] based on OrthoANI algorithm at EZBioCloud [106].

5.4. Experimental validation of probiotic propensities

For experiments, HK4 was inoculated in a sterile TSB medium and incubated at 37 °C (150 rpm). The late log phase cells were harvested by centrifugation at 8000 rpm for 10 min at 4 °C, washed twice using sterile phosphate buffer saline (pH 7.4) and re-suspended in the same to obtain dense cell suspension.

5.5. Acid and bile tolerance

For the determination of acid tolerance, the washed cells were inoculated in sterile TSB adjusted to different pH (pH 2 and pH 7) with the initial bacterial count of $\sim 10^7$ CFU/mL. The tubes inoculated with the cells were incubated at 37 °C for 90 min and the viable cell count was determined using the standard plate count method by plating 100 μ L of each serially diluted cell suspension on sterile TSA plates. The bile tolerance was determined by inoculating the washed cells in sterile 100 mL of TSB supplemented with 0.2% and 0.4% bile (HiMedia) to an initial count of $\sim 10^7$ CFU/mL and incubated at 37 °C for 24 h [107]. The media without bile supplementation was used as control. The viable count was determined employing standard plate count by plating 100 μ L of each serially diluted cell suspension on sterile TSA plates after 24 h.

5.6. Aggregation assay

For the auto-aggregation assay, the cell suspension with OD₆₀₀ of ~ 1.0 was dispensed in tubes, vortexed and incubated at 37 °C under static conditions. Absorbance (600 nm) was measured at 0 h, 2 h and 5 h interval by carefully withdrawing the supernatant. For the co-aggregation assay, the cell suspension of HK4 and the bacterial

pathogens namely *E. coli* and *S. enterica* with an optical density of 1.0 at 600 nm was prepared. Equal volumes of HK4 and pathogen cell suspension were mixed and vortexed thoroughly using a vortex mixer. Axenic bacterial suspensions were kept as controls. The tubes were incubated at 37 °C under static conditions and the absorbance (600 nm) was monitored at 0 h, 2 h and 5 h interval. Percentage auto-aggregation and co-aggregation were determined using the formula suggested by Collado and Meriluoto [30].

5.7. Cell surface hydrophobicity assay

The cell surface hydrophobicity of HK4 was determined based on its adherence to hydrocarbons [108,109]. The HK4 cell suspension of OD_{600 nm} ~ 0.5 was prepared. A 3 mL aliquot of washed cells was mixed with 1 mL of nonpolar organic solvents i.e. hexadecane, hexane and toluene, vortexed for 5 min and allowed to stand undisturbed for 30 min at room temperature (28 °C). Percent adherence of cells to the solvent was determined qualitatively based on the difference in the optical density of the aqueous phase at 600 nm before and after the assay.

5.8. Antibacterial activity of *Paenibacillus polymyxa* HK4

To demonstrate the antibacterial activity of HK4 against selected bacterial pathogens, an agar spot method was employed. The bacterial pathogens tested were *Aeromonas hydrophila* ATCC 49140, *Shigella sonnei* ATCC 25931, *Escherichia coli* O157:H7 ATCC 43888, *Listeria monocytogenes* BAA-751, *Vibrio parahaemolyticus* ATCC 17802, *Edwardsiella tarda* ATCC 15947, *Salmonella typhimurium* ATCC 13311, *S. enteritidis* ATCC 13076 and *S. typhimurium* ATCC 14028. An overnight grown culture of HK4 was spot inoculated at the center of sterile TSA plate and incubated at 37 °C for 24 h. The overnight grown pathogen suspension (100 μ L) was seeded in 4 mL of sterile 0.8% (w/v) soft agar and overlaid on plate pre-inoculated with HK4. The plate was further incubated at 37 °C for 24 h. The antibacterial activity of HK4 was recorded in terms of zone of inhibition to colony diameter ratio (ZOI/CD). Also, the exclusion of pathogens in liquid media (*E. coli* O157:H7 and *S. enteritidis*) by HK4 was evaluated using competitive exclusion broth assay employing protocol suggested by Ayala et al. (2019) [110] with several modifications. Briefly, HK4 and individual pathogen were co-inoculated in sterile TSB with initial CFU/mL of $\sim 10^7$ and incubated at 37 °C for 24 h. An aliquot of 0.5 mL was withdrawn, serially diluted and plated on TSA and MacConkey agar media (HiMedia, India) for enumeration of bacilli and pathogen, respectively at 10 h and 24 h interval.

5.9. Screening of HK4 for extracellular enzyme production

The production of extracellular enzymes namely amylase, protease, chitinase, cellulase, xylanase, pectinase and lipase by HK4 was determined using substrate hydrolysis method [3,111]. The isolate, HK4 was spot inoculated on basal salt agar plates supplemented with respective enzyme substrates viz. 1% (w/v) starch for amylase, 1% (w/v) casein for protease, 0.1% (w/v) colloidal chitin for chitinase, 1% (w/v) carboxymethyl cellulose for cellulase, 0.5% (w/v) corn cob xylan for xylanase, 0.5% (w/v) pectin for pectinase and 1% (v/v) tributyrin for esterase/lipase. The plates were then incubated at 37 °C for 24 to 48 h. To visualize the zone of substrate hydrolysis the starch containing plates were flooded with 0.1% iodine, casein containing plates with Frazier's reagent, chitin, xylan and CMC containing plates were flooded with 0.1% Congo red and pectin containing plates were flooded with 1% Cetyl trimethylammonium bromide (CTAB). The clear zone surrounding the colony signified the substrate hydrolysis owing to the production of extracellular enzyme.

5.10. Putative carbohydrate active enzymes in *Paenibacillus polymyxa* HK4

The carbohydrate utilization profile of HK4 was investigated using Hi-Carbo™ kit procured from HiMedia, India. For the experiment, HK4 was cultured in sterile TSB until the OD_{620 nm} reaches ~0.6. A cell suspension of 0.05 mL was aseptically inoculated in each well of the plate and the plate was incubated at 37 °C for 24 to 48 h. The results were recorded based on the color change observed and interpreted as per the information sheet provided in the kit. The putative genes associated with carbohydrate metabolism and catabolism of complex extracellular substrates were determined with the help of dbCAN2 metasever [112] (<http://cys.bios.niu.edu/dbCAN2>) and the enzymes associated with the pathways were annotated using KEGG database [91].

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygeno.2020.10.017>.

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